UE24CS352A / MACHINE LEARNING MINI-PROJECT

Predicting Eruptive Events at Volcanoes from Earthquake Data

A machine-learning study using the Kīlauea earthquake catalogue and Pu‘u ‘Ō‘ō eruption chronology

Team members: Mohit S and Nishanth N Aralaguppe

Demo platform: Google Colab

Abstract

This project investigates whether earthquake patterns can help identify volcanic eruptive activity and estimate time until an eruption begins. Using historical earthquake records from Kīlauea, Hawai‘i, and documented Pu‘u ‘Ō‘ō eruption episodes, we construct two supervised prediction tasks: binary classification of eruptive status and regression of days until the next eruption. Four approaches are compared: Logistic Regression, K-Means-based prediction, Random Forest and a neural network. Random Forest performs best on the held-out random test split, but a month-grouped validation experiment shows much weaker generalization. The results therefore support further investigation, not operational eruption warnings.

1. Problem Statement and Objectives

Volcanic unrest may coincide with changes in earthquake activity, but the relationship is not sufficient on its own to reliably forecast individual eruptions. The objective is to evaluate how much information earthquake location, depth, magnitude and recent earthquake frequency provide for identifying whether Kīlauea is erupting and for estimating time to the next eruption episode.

Task A — Classification	Task B — Regression
Predict erupting (1) or repose (0) at the time of an earthquake.	Estimate days until the next episode starts; assign 0 when the volcano is already erupting.

2. Dataset and Features

The earthquake catalogue contains 6,485 events from 1 January 1983 to 30 December 1986, sourced from the Advanced National Seismic System (ANSS) catalogue via WOVOdat. The eruption table contains 49 Pu‘u ‘Ō‘ō episodes from the Hawai‘i Center for Volcanology chronology. A geographic East Rift Zone filter reduces the earthquake set to 3,764 records. Labels identify 844 erupting records (22.4%) and 2,920 repose records (77.6%).

Feature group	Inputs used	Purpose
Event properties	Latitude, longitude, depth, magnitude	Describe each earthquake’s location and size.
Recent activity	Counts in previous 1, 7 and 30 days	Represent short- and medium-term earthquake rate.

The labels are derived from eruption start dates and durations. A 24-hour buffer is added to episode ends because the start dates have day-level precision. For earthquake-rate features, only earlier earthquakes are counted to avoid using future events as inputs.

3. Methodology

1. Prepare	2. Engineer	3. Train	4. Evaluate
Load and sort records; filter to the East Rift Zone; build target labels.	Calculate earthquake rates over prior 1-, 7- and 30-day windows.	Compare Logistic Regression, K-Means, Random Forest and neural network models.	Use held-out data and examine generalization with month-grouped folds.

The data are split into training (70%), development (20%) and test (10%) subsets using stratification by eruptive status. The feature scaler is fit on training data only. Hyperparameters are selected using the development set. For classification, AUROC and Cohen’s Kappa are reported because the labels are imbalanced; regression is assessed with RMSE in days and R^2.

RESULTS / INTERPRETATION

4. Results and Discussion

Metrics below are taken from the fresh Google Colab results package generated on 13 October 2026. Test-set metrics are reported unless the table explicitly says cross-validation.

Task A — Classification of eruptive status

Model	Test AUROC	Test Kappa	Recall	Precision
Logistic Regression	0.68	0.20	0.19	0.59
K-Means-based	0.64	0.13	0.55	0.29
Random Forest	**0.83**	**0.37**	**0.32**	**0.75**
Neural Network	0.69	0.21	0.17	0.78

Random Forest gives the strongest held-out classification performance (AUROC 0.83). Its test confusion matrix is TN=284, FP=9, FN=57 and TP=27. Thus it misses many erupting cases at the selected classification threshold, despite achieving the best AUROC. Accuracy alone is not a suitable headline metric because 77.6% of records are labelled repose.

Task B — Regression of time until eruption

Model	Test RMSE	RMSE in days	Test R^2
K-Means-based	1.35 (scaled)	18.1	−0.70
Random Forest	**0.76 (scaled)**	**10.1**	**0.47**
Neural Network	0.88 (scaled)	11.8	0.28

Random Forest also performs best on regression, with a test RMSE of approximately 10.1 days and R^2=0.47. The 30-day earthquake rate is the largest reported feature-importance contribution for this model (0.41), followed by the 7-day rate (0.16). Feature importance describes the fitted model, not a causal effect.

Key limitation — performance falls under time-grouped validation. In 5-fold cross-validation, random folds give AUROC 0.83, Kappa 0.46 and regression R^2 0.39. When folds are grouped by calendar month, these fall to AUROC 0.58, Kappa 0.03 and R^2 −0.17. Similar earthquake swarms can appear in both random training and test folds; grouping by month is a stricter test of generalization to different time periods. This result limits any claim that the model is ready for real-world eruption forecasting.

5. Conclusion and Future Work

On the random held-out split, Random Forest is the strongest of the tested methods for both tasks. However, lower recall for erupting cases and the substantial drop under month-grouped validation show that the learned patterns may not transfer reliably to new time periods. The system should be treated as an exploratory machine-learning study, not an operational warning tool. Future work should test longer and more varied time periods, add independent signals such as ground deformation and gas emissions, and use temporally separated evaluation before drawing forecasting conclusions.

Implementation and reproducibility

The project is implemented in a Google Colab notebook with Python, pandas, NumPy, scikit-learn, Matplotlib and PyTorch. The notebook builds the features and targets from the two source CSV files and saves tables, plots and metadata from the same run. The seed is set to 42; model behaviour can still vary slightly across software/hardware environments, particularly for neural networks.

Project repository: github.com/mohit2007malur-png/volcano-eruption-prediction

References

[1] B. Mullet, A. Singh and Y. Q. Wong, “Predicting Eruptive Events at Volcanoes from Earthquake Data,” Stanford CS229 Machine Learning project, 2019. Code and paper referenced by the supplied project materials.
[2] Advanced National Seismic System (ANSS) earthquake catalogue, accessed via WOVOdat; historical Pu‘u ‘Ō‘ō eruption chronology, Hawai‘i Center for Volcanology. Dataset notes and links are included in the project code archive.